THE RATTY2TAILS PAPERS

Quantitative Easing

The emergency tool that became the default setting

Ratty2Tails

MONETARY HISTORY SERIES

From a Tokyo experiment in 2001 to a permanent fixture of the toolkit

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Quantitative Easing

How an emergency experiment in Tokyo became the default setting for every major central bank

TL;DR

- Quantitative easing (QE) is a central bank creating new bank reserves to buy financial assets — usually government bonds — directly from the private sector, typically used once short-term interest rates are already near zero and can't be cut much further.
- The tool and its name were invented by the Bank of Japan in 2001 as an emergency deflation-fighting experiment and lifted in 2006; almost nobody expected it to become the default response to every major Western financial shock for the next quarter-century.
- Between 2008 and 2022, the Federal Reserve, the Bank of England and the European Central Bank expanded their balance sheets from a combined few trillion to a combined peak of roughly $20 trillion-equivalent, buying trillions of pounds, euros and dollars of bonds along the way.
- The best empirical estimates suggest QE lowered long-term borrowing costs by tens of basis points at a time — not that it transformed growth or inflation outright — and even that modest effect is disputed and hard to isolate from everything else happening during a crisis.
- Unwinding QE turned out to be far harder than deploying it: the Fed's tightening programme ended in December 2025 having reversed only about half of the pandemic-era expansion, and within days the Fed announced it would start buying bonds again to keep bank reserves "ample" — with inflation still running above target.
- A tool introduced as a temporary, unconventional emergency measure now looks like a permanent fixture of the monetary toolkit — one central banks can seemingly switch on far more easily than they can switch off.

INTRODUCTION & SCOPE

The companion essay in this series, *The Cantillon Effect*, asked who benefits when a central bank creates new money and why. This essay asks a narrower, more mechanical question about the specific tool that has done almost all of that money creation for the past two and a half decades: what quantitative easing actually is, where it came from, whether it works, and why — twenty-five years after a slightly desperate experiment in Tokyo — the world's largest central banks appear to have concluded they can never quite put it down again.

Quantitative easing has a peculiar biography. It was born as a niche, faintly embarrassing response to a uniquely Japanese problem: a decade of stagnation and deflation that conventional interest-rate policy, already pinned at zero, could not touch. For years afterwards it was treated by most Western central bankers as a curiosity — something the Bank of Japan had tried because it had run out of better options, not a tool any serious central bank would reach for by choice. Then, in the autumn of 2008, with the global financial system in freefall and interest rates already near the floor, the Federal Reserve reached for it anyway. So, in short order, did the Bank of England, and — more cautiously — the European Central Bank. What had been an obscure Japanese experiment became, within a few years, the default global response to any sufficiently large economic shock: the 2008 financial crisis, the eurozone sovereign debt crisis, and most dramatically the 2020 pandemic, each met with rounds of central bank bond-buying larger than the last.

This essay traces that arc in full: what QE is and how it differs from ordinary monetary policy; its origins in Japan between 2001 and 2006; its global adoption during the 2008 crisis and its extraordinary expansion during the pandemic; the channels through which it is supposed to affect the economy and what the evidence says about whether those channels actually work; the difficult and still-unfinished business of unwinding it through quantitative tightening; and the remarkable, very recent turn in which the Federal Reserve — having just finished shrinking its balance sheet — announced in December 2025 that it would start growing it again.

SECTION ONE

What Quantitative Easing Actually Is

Strip away the acronyms and quantitative easing is a simple transaction repeated at an enormous scale: a central bank creates new reserves — balances that only it can create, and that only banks and a handful of other institutions are allowed to hold — and uses them to buy financial assets, overwhelmingly government bonds, from banks, pension funds, insurance companies and other institutional investors. The seller receives newly created central bank money in exchange for a bond it used to hold; the central bank's balance sheet grows by the value of the bond on one side and the new reserves on the other.

This is different from ordinary monetary policy in an important way. Central banks normally influence the economy by setting a short-term interest rate — the rate at which banks lend to each other overnight — and letting that rate ripple out through the rest of the financial system. QE is what a central bank does once that short-term rate is already close to zero and cannot usefully be cut any further: rather than making short-term borrowing cheaper still, it tries to push down longer-term borrowing costs directly, by buying the longer-dated bonds whose yields set the price of mortgages, corporate credit and government debt itself. It is also, in a narrower technical sense, different from simply "printing money" in the everyday sense people often mean by that phrase: QE swaps one financial asset (a bond) for another (a reserve deposit) with an institution that already held the bond; it does not, by itself, hand newly created money to a household or a business, though the companion essay in this series explains why that distinction matters less than it might seem once the new money starts circulating.

SECTION TWO

Origins: Tokyo, 2001–2006

The term "quantitative easing" and the policy it describes were both invented by the Bank of Japan. By the start of the twenty-first century, Japan had spent a decade mired in stagnation and mild deflation following the collapse of its late-1980s asset bubble, and the Bank of Japan had already cut its overnight call rate — its equivalent of the federal funds rate — to zero. With conventional policy exhausted, the Bank, under Governor Masaru Hayami, announced on 19 March 2001 that it would change its operating target altogether: instead of targeting the overnight interest rate, it would target the outstanding balance of current accounts that commercial banks held with it, deliberately pushing that balance far above the level banks were required to hold. It set an initial target of roughly ¥5 trillion, up from about ¥4 trillion, and committed to maintain the policy until core consumer prices were rising "stably" at zero per cent or above.

Over the following years the Bank of Japan raised that target repeatedly — nine times in total — until the current-account balance reached ¥35 trillion by December 2004. At the time, plenty of economists were openly sceptical that any of this would matter: with interest rates already at zero, flooding banks with yet more reserves that paid no interest looked, to many observers, like swapping one zero-yielding asset for another and expecting a different result. When the Bank of Japan finally lifted the policy in March 2006, having judged that mild deflation had given way to stable, if barely positive, inflation, the retrospective verdict from its own economists and outside researchers was mixed: the policy appeared to have helped stabilise a fragile banking system and to have modestly shifted expectations about future interest rates, but it had done comparatively little to stimulate the real economy on its own, doing more once financial stability had already been restored than it did to restore that stability in the first place.

> **ASIDE — AN IDEA BEFORE ITS TIME**
>
> Japan's experience mattered enormously a few years later for a reason nobody in Tokyo could have anticipated: it gave Western central bankers a working precedent — however imperfect — to point to when they, too, ran out of room to cut interest rates. Ben Bernanke, who became Federal Reserve Chair in 2006, had studied Japan's lost decade closely as an academic economist; when the 2008 crisis hit, the Fed did not have to invent quantitative easing from scratch. It had a case study, even if that case study came with a decidedly qualified verdict attached.

SECTION THREE

Going Global: The 2008 Crisis and the Trillion-Dollar Decade

The 2008 financial crisis turned quantitative easing from a Japanese curiosity into the default emergency setting for the entire developed world. The Federal Reserve began large-scale asset purchases in November 2008, initially targeting mortgage-backed securities and agency debt at the epicentre of the crisis, and followed with further rounds — popularly numbered QE1, QE2 and QE3 — that extended into outright Treasury purchases and ran, in one form or another, until late 2014. The Bank of England began its own asset purchase programme in March 2009, buying gilts and a smaller quantity of corporate bonds; over the following decade its holdings would grow, in stages, to a peak of roughly £895 billion. The European Central Bank moved more cautiously and by different means at first, relying heavily on long-term loans to banks rather than outright bond purchases, before eventually launching its own large-scale Asset Purchase Programme in 2015 and expanding it steadily thereafter.

What began as a series of one-off emergency interventions hardened, over the following decade, into a semi-permanent feature of monetary policy — deployed again during the 2011–12 eurozone sovereign debt crisis, tapered cautiously from 2013 to 2014, and then, in March 2020, redeployed at a scale that dwarfed everything that had come before. In a matter of weeks, the Federal Reserve purchased more Treasury and mortgage-backed securities than it had in years of the post-2008 programmes combined; the Bank of England and the European Central Bank moved just as fast, the latter launching a dedicated Pandemic Emergency Purchase Programme that would eventually total some €1.7 trillion. By the time the dust had settled in 2022, the combined balance sheets of the Federal Reserve, the Bank of England and the European Central Bank had reached a scale that would have seemed like fantasy to the central bankers who first eyed Japan's experiment warily in the early 2000s: the Fed's balance sheet alone peaked at just under $9 trillion in the spring of 2022, and the ECB's at just over €8.8 trillion in the summer of that year.

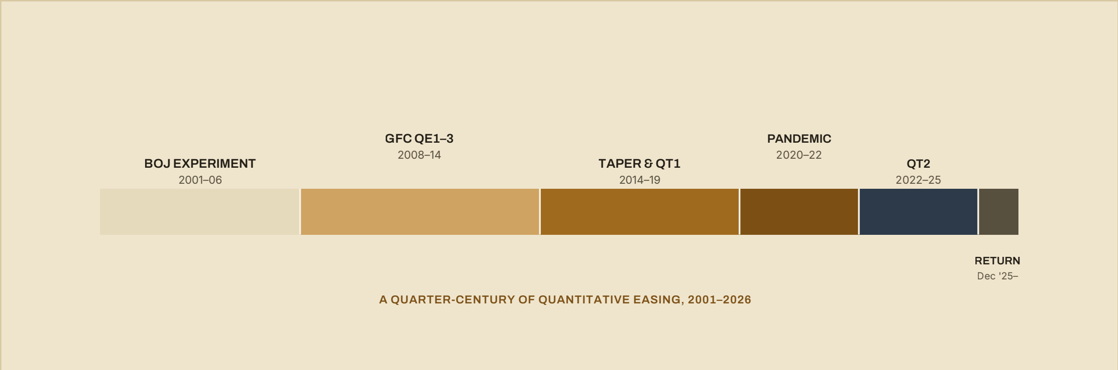

Figure 1. The major phases of quantitative easing, from the Bank of Japan's original experiment to the Federal Reserve's December 2025 return to asset purchases. Segment widths are approximately proportional to duration; dates are approximate era boundaries, not precise policy start and end dates.

SECTION FOUR

How It's Supposed to Work: the Transmission Channels

Central banks and academic economists generally point to several distinct mechanisms through which buying large quantities of bonds is supposed to affect the wider economy, and it is worth being precise about them, because "QE works" and "QE doesn't work" are both too vague to be useful without specifying which channel is under discussion.

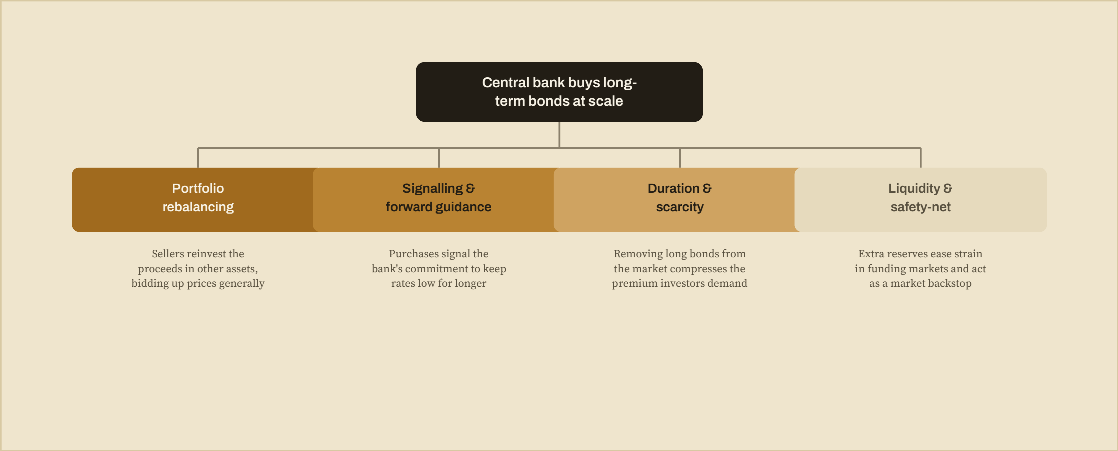

Figure 2. The four channels most commonly cited for how large-scale asset purchases are meant to affect financial conditions. In practice these channels overlap and reinforce one another, and disentangling their individual contributions is one of the harder problems in empirical monetary economics.

The **portfolio-rebalancing** channel is the most intuitive: when a central bank buys bonds from a pension fund or insurer, that institution does not simply sit on the cash — it reinvests the proceeds in other assets, which pushes up the price of everything from corporate bonds to equities to property, and in doing so lowers the cost of borrowing across the economy, not just for the government. The **signalling** channel works through expectations rather than transactions: a large asset-purchase programme is also a credible signal that the central bank intends to keep short-term rates low for an extended period, which itself pulls down longer-term rates that are, in part, an average of expected future short-term rates. The **duration and scarcity** channel is more technical: by removing a large quantity of long-dated bonds from the market, a central bank reduces the amount of interest-rate risk the private sector has to hold in aggregate, which in standard asset-pricing models should lower the extra yield investors demand for holding it. And the **liquidity and safety-net** channel — the one research by Valentin Haddad, Alan Moreira and Tyler Muir has emphasised most recently — captures something subtler still: once markets learn that a central bank will reliably step in to buy government debt during a crisis, investors may treat that debt as safer than they otherwise would, for reasons that have little to do with any single purchase and everything to do with the standing commitment behind it.

SECTION FIVE

Does It Work? What the Evidence Shows

The honest answer is: probably, somewhat, and less predictably than anyone would like. The most-cited empirical estimates come from studies of the Federal Reserve's early rounds of purchases between 2008 and 2011. Working papers from Swiss National Bank researchers Nikola Mirkov and Barbara Sutter, and separately Signe Krogstrup, Samuel Reynard and Barbara Sutter, estimated that the cumulative effect of quantitative easing on ten-year US Treasury yields during that period was on the order of 46 to 85 basis points from a liquidity effect alone, plus a further 20 basis points or so from the separate portfolio-rebalancing effect of removing bonds from the market. These are not trivial numbers — the equivalent of a moderate interest-rate cut applied directly to long-term borrowing costs — but they are also a long way from suggesting that QE single-handedly rescued the post-2008 recovery, and estimates from other studies, using different methods and time periods, range both above and below this band.

More recent research has approached the question from an unexpected angle: rather than asking how much QE moved yields on any given day, Haddad, Moreira and Muir examined how QE appears to have changed the underlying relationship between government debt and its perceived riskiness. Before 2008, in a large sample of developed economies, a country whose debt-to-GDP ratio rose by ten percentage points would typically see its long-term borrowing costs rise by around 12 basis points, consistent with investors demanding a higher return for taking on more sovereign risk. After the adoption of quantitative easing, that relationship weakened substantially, and in some episodes reversed entirely — government debt has, in a sense, come to be viewed by markets as safer than the underlying fiscal arithmetic alone would suggest, because a large and willing buyer of last resort now reliably stands behind it during periods of stress.

On the harder question of whether QE meaningfully raised growth or inflation, the record is murkier still, and Japan's own experience is a useful corrective to over-confidence in either direction. The Bank of Japan's own retrospective assessments of its 2001–2006 programme found that it had helped stabilise a fragile banking system but had done comparatively little to lift growth or inflation directly until that stabilisation had already taken hold — suggesting that QE may function less as a general-purpose growth lever and more as a targeted tool for repairing broken financial plumbing, with its effects on the wider economy following indirectly, if at all, once the plumbing is fixed.

SECTION SIX

The Unwind: Quantitative Tightening and Its Discontents

If deploying quantitative easing turned out to be relatively straightforward — announce a purchase programme, and buy — unwinding it has proven considerably harder in practice. The reverse process, quantitative tightening (QT), can be done passively, by simply not reinvesting the proceeds of maturing bonds and letting the balance sheet shrink on its own, or actively, through outright sales; central banks have tried both, with mixed results.

The Federal Reserve's first attempt at QT ran from 2017 to 2019 and ended abruptly when funding markets seized up in September 2019, forcing the Fed to resume growing its balance sheet within months, well before the pandemic gave it a further reason to do so. Its second, much larger attempt began in June 2022 from a peak of roughly $8.97 trillion and ran until December 2025, by which point the balance sheet had fallen to around $6.7 trillion — a reduction of some $2.3 trillion, but, on the Fed's own reckoning, only about half of the increase added during the pandemic. The Bank of England took a more assertive approach, becoming the first major central bank to sell bonds outright rather than simply letting them mature, reducing its gilt and corporate-bond holdings from a 2021 peak of roughly £895 billion by roughly a third to a half over the following several years — a process punctuated, memorably, by the September 2022 crisis in the UK government bond market, when a bout of gilt-market turmoil forced the Bank to pause its sales and briefly buy long-dated gilts again to restore order, a vivid illustration of how quickly the tension between shrinking a balance sheet and preserving financial stability can resurface. The European Central Bank's approach has been the most passive of the three, unwinding both its bond portfolio and its emergency loans to banks mainly through non-reinvestment and voluntary early repayment, taking its balance sheet down from a peak of roughly €8.8 trillion in mid-2022 to around €6.1 trillion by the end of 2025.

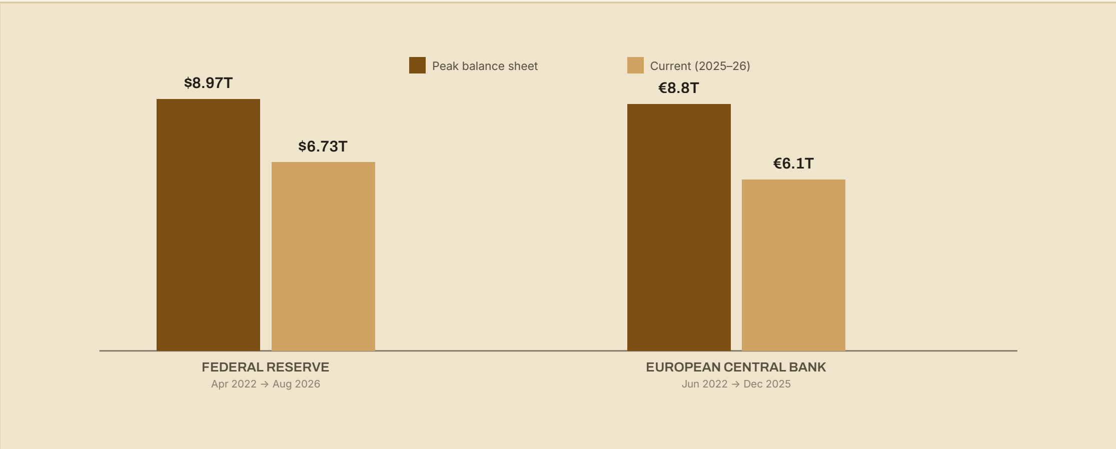

Figure 3. Peak versus most recently reported total assets, Federal Reserve and European Central Bank. Sources: Federal Reserve H.4.1 release via FRED (WALCL); ECB weekly financial statements, as reported by CPRAM and Wolf Street. The Bank of England's gilt and corporate-bond holdings followed a broadly similar arc, from a 2021 peak of roughly £895 billion.

SECTION SEVEN

December 2025: Quantitative Easing Returns

On 10 December 2025, days after formally declaring an end to the largest quantitative tightening programme in its history, the Federal Reserve announced that it would begin buying assets again — reportedly around $40 billion a month to start — in order to maintain what the Fed calls "ample reserves" in the banking system. Officially, this is not a rebranded return to crisis-era QE: it is presented as a technical operation to keep bank reserves comfortably above the minimum level needed for smooth-functioning money markets, the same kind of plumbing concern that forced the Fed's hand during the September 2019 repo market disruption, rather than an attempt to stimulate demand or lower long-term borrowing costs. Skeptics have been quick to note the awkward timing: consumer prices were, at the time of the announcement, still rising at around 3 per cent a year, comfortably above the Fed's 2 per cent target, and the balance sheet — even after three and a half years of tightening — remained roughly $2.4 trillion, or around 60 per cent, larger than it had been at the end of 2019. Whatever the technical justification, the practical result is that a balance sheet which had only just finished shrinking began growing again within days, having reversed only about half of what the pandemic had added to it in the first place.

Whether this particular episode should be called "QE" is partly a semantic argument, and reasonable people can disagree about how much weight to put on the Fed's own framing versus its practical effects. What is harder to dispute is the pattern it fits into. A tool invented in 2001 as a last-resort experiment for a single stagnant economy has, in the space of a quarter-century, been used by every major Western central bank, at ever-larger scale, in response to ever more varied kinds of shocks — a banking crisis, a sovereign debt crisis, a pandemic, and now, apparently, simply the ordinary business of keeping reserves at a level central banks consider comfortable. Each time a tightening cycle has approached anything resembling the pre-2008 size of these balance sheets, something — a repo market seizure, a pandemic, a reserve-scarcity concern — has intervened before it got there. It remains an open question, and one this series will return to, whether that pattern reflects bad luck in the timing of shocks, or whether a financial system that has spent most of two decades operating with vastly larger central bank balance sheets has, in some more fundamental sense, come to depend on them.

SUMMARY

In Short

Quantitative easing began life as a narrow, faintly desperate experiment: a way for the Bank of Japan to keep fighting deflation after it had already cut interest rates to zero and run out of conventional options. Between 2008 and 2022, that experiment was adopted, refined and repeatedly scaled up by the Federal Reserve, the Bank of England and the European Central Bank, taking their combined balance sheets from a few trillion dollars-equivalent to a peak of roughly twenty trillion. The evidence suggests it works, in the narrow sense of lowering long-term borrowing costs by a meaningful but not enormous amount and of changing how markets price government debt — but it is not a lever that reliably or predictably drives growth and inflation on its own, and its clearest successes have come alongside efforts to stabilise a broken financial system rather than in place of them.

What is most striking, a quarter-century on, is how difficult the tool has proven to put down once picked up. Every attempt to shrink these balance sheets back toward their pre-2008 size has been interrupted — by a repo market disruption in 2019, a pandemic in 2020, and a reserve-scarcity concern in December 2025 that saw the Federal Reserve resume asset purchases within days of finishing three and a half years of tightening, having reversed only half of what it had added since 2020. Whether that is a sign of bad luck, of a financial system that has grown structurally dependent on ample central bank reserves, or simply of the fact that shrinking a balance sheet is inherently harder than growing one, is not yet settled. What does seem settled is that the emergency measure of 2001 is no longer an emergency measure at all: it is part of the ordinary furniture of modern central banking, and the companion essay in this series on the Cantillon effect explains why that matters for more than just central bankers' balance sheets.

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